

Comparative Analysis of Face Emotion Detection based on Deep Learning Techniques

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Abstract— This research paper presents a comparative analysis of emotion recognition using deep learning techniques. The aim of the study was to compare the performance of state-of-the-art deep learning models, namely Convolutional Neural Network (CNN), Mobilenet and Long Short-Term Memory (LSTM). The dataset used for our work is the popular FER - 2013 dataset, which consist of annotated tweets with emotion labels. The proposed work evaluates the models based on their precision, accuracy, recall, and F1-score. The results show that the CNN model, which was trained using the image dataset, outperforms the other models in terms of precision and F1-score. The study also analyzes the effect of various pre-processing techniques on the performance of the models. Overall, the study provides a comprehensive analysis of emotion recognition using deep learning models and highlights the strengths and weaknesses of different approaches. The results of this study are useful for researchers and practitioners working in the fields of natural language processing and emotion recognition.

Keywords: *Emotion Detection, Comparative Analysis, Deep Learning, Feature Extraction, Facial Expression Recognition, Performance Analysis.*

I. INTRODUCTION

Nowadays emotion detection plays a key part in the research area, with many applications. Emotion detection is about identifying the emotions of humans like anger, sadness, happiness, neutrality, depression, and many more using advanced techniques like deep learning [1]. Deep learning techniques can be used to analyze facial expressions and emotion detection. In general, facial expressions play a vital role in identifying emotions but there are also other ways to detect emotions like through speech, physiology, audio, or video [2]. There are various applications of emotion detection in AI, including customer service, market research, and mental health. For example, companies can use emotion detection to analyze feedback from customers and improve their products and services based on feedback. In mental health, emotion detection can be used to monitor and diagnose mental health conditions by analyzing a person's speech patterns, facial expressions, and other physiological data and knowing if the person is depressed or not [3]. It can also be used to recommend music, food, and videos based on

a person's mood. Emotion detection plays a vital role in humanoid robots making them efficient to understand a person like a human by detecting their emotions and responding to them as required.

Deep Learning is one of the most popular machine learning practices in which computers are taught to perform specific tasks that are natural for humans [4]. A basic example of helping you get a better understanding of deep learning technology may have voice control use in devices such as hands-free speakers, phones, tablets, and TVs. It is behind a variety of artificial intelligence apps and services that can help improve the automation and performance of many physical and analytical tasks without human intervention [5].

Non-verbal cues, such as facial expressions, can be used to infer a person's emotional or mental condition. Facial expression recognition is the process of identifying human emotions from observed facial expressions. Humans have an innate talent for understanding emotions based only on facial expressions [6]. Even infants as young as 36 hours old can read basic emotions in people's faces. Among the senior population, this skill ranks high on the list of vital social competencies. Only a small subset of human emotions can be conveyed through facial expressions in a form that is understood by everyone. A well-trained system can understand emotions independently of the subject; hence this universality might be useful in human-machine interaction if modelled accurately [7]. Computerized face expression recognition has numerous practical uses. Analysis of facial expressions and subsequent identification of human emotions adds a new level to human-machine interactions [8]. Some consumer digital cameras and interactive ads use smile detection like this. Robots may benefit from software that can recognize human emotions. If robots can accurately predict human emotions, they can act in accordance with those predictions [9].

Every person is a vulnerable target for the persuasive power of their emotions. People express their feelings in several different ways, including through their voices, bodies, and facial expressions [10]. Facial expressions are a crucial aspect of human communication even though they are wordless. One or more of the superficial muscles of the face

are responsible for facial expression when they are contracted. Facial expressions are the outward manifestations of a person's inner emotional state, intent, or communications. It's the clearest and most telling sign of how someone's mind is feeling. The field of FER is crucial to the study of human-computer interaction. With the help of facial expression recognition software, a computer visualization system can communicate with humans.

II. LITERATURE SURVEY

Li J et al., [1] expressed that a micro-expression is a face movement that is discrete, localised, and fleeting. It is regarded as a key hint for lie detection since it has the ability to expose the true feelings that a person is trying to hide from another person. The research on micro-expression has garnered a lot of attention because of the numerous potentials it offers in a variety of sectors. Micro-expression recognition, on the other hand, is fraught with difficulty because of the brief length and low intensity of the movements associated with micro-expressions; this means that accuracy still has room for improvement. The authors suggest a deep local-holistic network method for recognising micro-expressions. Psychological research on how much focus is put on different facial expressions inspired this approach. The goal of the study was to increase the effectiveness of the extraction of micro-expression features. The suggested method includes two sub-networks, one for obtaining local characteristics and the other for obtaining global features. The combined sub-networks are used to perform micro-expression recognition with the help of the effective features that were retrieved and extracted. The authors tested the proposed strategy on the datasets that are the most widely utilised. The findings of the experiments indicate that the proposed strategy achieves a performance that is satisfactory overall.

Ali I. Siam et al., [2] presented that one of the scientific disciplines that is now on the rise is emotion recognition. It is utilised in a variety of different contexts. Robotic vision and interactive communication between robots are two of the most exciting uses for this technology. Both oral and visual modalities can be used to decipher an individual's emotional state. Expressions on a person's face are often believed to be the most reliable indicator of their emotional state. The work that is being proposed outlines a real-time strategy for putting emotion recognition into practise and putting it to use in robotic vision applications. The strategy under consideration is divided into four stages. In addition, principal component analysis is used to carry out the process of feature decomposition. This phase is implemented to improve the reliability of the emotion detection process. At long last, the dissected features are incorporated into a machine learning strategy. In addition, the authors used a Multilayer Perceptron, often known as an MLP, which is a method for creating effective deep neural networks. The methodologies that are described are then tested on a variety of datasets using a variety of evaluation metrics. The results of the simulation show that they obtain an outstanding performance with a human emotion identification accuracy of 97%, which ensures that they attain superiority among the efforts made in this field.

Naveen Kumar H N et al., [3] discussed that the non-verbal character of face expression makes it one of the earliest forms of communication, predating vocal

communication in both its roots and its conceptualization. Most of the currently available methods for AFER concentrate primarily on the extraction of global facial features on the assumption that each area of the face contributes an equal amount of information that can be utilised to discriminate between different expressions. Not enough research has been done on how to discover and localise key face areas for expression identification, or how to derive highly discriminative characteristic distribution from these regions. Clearly, there is space for development here. The proposed work made several important contributions, including developing an innovative feature distribution through integrating the ability to discriminate of shape and texture features, determining the impact of facial regions, and identifying popular facial regions which hold abstract and highly biased data for expression recognition. The CK+, KDEF, and JAFFE benchmark facial expression datasets have been utilised in the work that has been proposed. The proposed approach has a much higher recognition rate in comparison to the handmade feature-based methods that are already in use.

K. Mohan et al., [4] illustrated that since a picture is worth a thousand words, a photo of a person's face conveys a wealth of information about their appearance, identity, age, and mood. Expressions of emotion on the face are frequently analysed for their behavioural significance in social interactions. A difficult job in the field of computer vision, automatic facial expression recognition has many potential uses in fields as diverse as driver safety, human-computer interface, healthcare, behavioural science, video conferencing, cognitive research, and many more. In this paper, we present a deep learning-based method for reading a person's emotions from their face. There are two stages to the suggested procedure. The first uses a local gravitational force descriptor to extract characteristics from facial photos, and the second uses the descriptor as input into a unique deep convolutional neural network (DCNN) model. Five benchmark datasets are used to test the proposed method with existing methods. There are seven primary feelings represented in the data sets. Four criteria are used to compare the proposed method to other methods already in use. The proposed technique has been shown to outperform current technologies in all tested datasets.

P. Rodriguez et al., [5] expressed that the unpleasant sensation of pain has been proven to have a significant role in the healing process. There is a need for automated solutions to measure this since it requires a lot of time and money to manually assess. Instead of relying solely on facial features, as is the case with current state-of-the-art pain assessment methods, the authors of this paper propose feeding raw frames to deep learning models in order to improve performance. This approach not only outperforms the most recent state-of-the-art results, but also directly tackles the issue of imbalanced data. To begin, we use VGG Faces to train convolutional neural networks (CNNs) to recognise faces, and then we couple these CNNs with a long short-term memory to take advantage of the time-based relationship between video frames. Moreover, we contrasted the results obtained by utilising the widely-appreciated paradigm based on the canonically normalised appearance to those obtained by considering the whole picture. Therefore, findings acquired on different databases should be surpassed.

III. METHODOLOGY

This section explains datasets [8] and techniques used in our work. Figure 1 depicts the workflow of the proposed work.

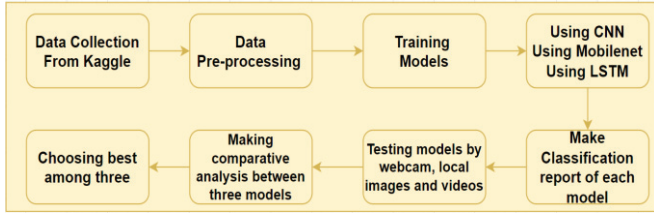


Fig. 1. Workflow Diagram

TABLE I. DATASETS INFORMATION

SL.NO	CATEGORY	NO.OF. IMAGES IN TRAIN	NO.OF. IMAGES IN TEST
1	Neutral	4965	1233
2	Happiness	7215	1774
3	Surprise	3171	831
4	Sadness	4830	1247
5	Anger	3995	958
6	Disgust	436	111
7	Fear	4097	1024

Table 1 provides the information related to categories of image. Total number of images are used for training and testing. In the context of training Convolutional Neural Network (CNN) models, data pre-processing plays a vital role in achieving accurate and reliable results. Pre-processing techniques such as data enhancement, normalization, and resizing are commonly used to prepare image datasets for use in CNN models. Data augmentation techniques involve generating new training models by applying transformations such as rotation, translation, and rotation to the original images. Normalization helps center and scale the pixel values of images, making them easier to work with. Resizing is used to ensure that all images are the same size, which is required to render them in the CNN model.

A. Convolutional Neural Network

CNN is a type of neural network commonly used in computer vision tasks such as image recognition and object detection [11]. CNNs are designed to detect patterns in visual data by using filters that convolve over the input image, allowing the network to learn features at different levels of abstraction. The main components of CNN are convolutional layers, pooling layers, and fully connected layers. Convolutional layers use filters to detect features in the input image, and pooling layers reduce the size of feature maps, making them more computationally efficient. Fully connected layers are used for classification, taking features learned by convolutional and pooling layers and outputting a prediction for the input image.

B. Mobilenet

MobileNet is a popular neural network architecture designed specifically for mobile devices with limited processing power and memory. It is developed by Google and introduced in 2017. It is a lightweight neural network designed to be efficient in terms of both computational

complexity and memory consumption. It is commonly used for tasks such as object recognition and image classification.

C. Long short-term memory (LSTM)

It is a type of recurrent neural network used for sequence modeling and prediction [12]. LSTMs solve the vanishing gradient problem that occurs in traditional RNNs and are capable of learning long-term dependencies in sequential data. LSTMs use memory cells and control gates to select information from the past to remember or forget, and to capture complex patterns and dependencies in sequential data. LSTMs have been used for speech recognition, natural language processing, and time series prediction, among other tasks.

D. Training Model

Training deep learning models such as convolutional neural networks (CNNs), long short-term memory (LSTM) networks, and mobile nets involves feeding large amounts of labeled data through the network to adjust parameters and weights to learn underlying patterns. Information [13]. The training process consists of two main steps: forward propagation and backpropagation. In the forward propagation step, the input data is fed through the network layers and the output is compared with the true labels using a loss function. In the backpropagation step, the error is propagated backward through the network layers and the weights are updated using gradient descent to minimize the loss function.

E. Classification Reports

When evaluating the performance of a trained classification model, the classification report is a useful tool that summarizes the precision, recall, and F1 score for each class [14]. To generate a classification report, first need to make predictions on the test set using trained model. Then, comparison is made on predictions to the test set's true labels to calculate various metrics for each class. Precision measures how many of the predicted positive samples are positive, while recall measures how many true positive samples are correctly predicted. The F1 score is the harmonic mean of precision and recall and provides an overall measure of model performance. By examining the classification report, easy to identify which model is performing well in, and which ones need further improvement.

F. Testing Models

Testing machine learning models through live prediction, local images and videos is a key step in the development process to assess their accuracy and performance. Live prediction involves feeding a model with real-time data to make predictions on the fly, such as predicting the class of an object in a camera feed. Local image testing is the evaluation of model performance on a set of images stored on a local device. Video testing involves feeding a model with a series of images to make predictions about identifying objects or events in a video.

IV. RESULT AND DISCUSSIONS

In this section the performance metrics are measured for CNN, MobileNet and LSTM models. The proposed models are tested on benchmarked datasets.

A. Comparative analysis

A comparative analysis between three models involves evaluating their performance on a given task to determine which model is the most effective. Analysis usually involves measuring various metrics such as precision, accuracy, recall and F1 score. These metrics provide insight into how well the models are performing and where they excel or fall short. In addition, comparative analysis may involve identifying factors that contribute to the structure, parameters, and hyperparameters of each model. By comparing the results of models, researchers and practitioners can better understand which methods and structures are most effective for a given problem and use that knowledge to improve their models in the future. Table 2 depicts the accuracy values of the proposed models. The accuracy of the proposed models such as CNN, Mobilenet and LSTM are 97%, 87% and 82% respectively.

TABLE II. COMPARATIVE ANALYSIS

MODEL NAME	ACCURACY	MACRO AVG	WEIGHTED AVG
CNN	0.97	0.96	0.97
MOBILENET	0.87	0.87	0.87
LSTM	0.82	0.77	0.77

The performance metrics of various emotions are analyzed using CNN model. Table 3 shows the performance analysis values and same are depicted in Figure 2.

TABLE III. PERFORMANCE ANALYSIS OF CNN MODEL ON IMAGE DATASETS

	PRECISION	RECALL	F1-SCORE	SUPPORT
Angry	0.95	0.95	0.95	3995
Disgusted	0.92	0.9	0.91	436
Fearful	0.96	0.96	0.96	4097
Happy	0.99	0.99	0.99	7215
Neutral	0.97	0.97	0.97	4965
Sad	0.97	0.97	0.97	4830
Surprised	0.96	0.96	0.96	3171
Accuracy			0.97	28609
Macro avg	0.96	0.96	0.96	28609
Weighted avg	0.97	0.97	0.97	28609

The performance metrics of various emotions are analyzed using Mobilenet model. Table 4 shows the performance analysis values and same are depicted in Figure 3.

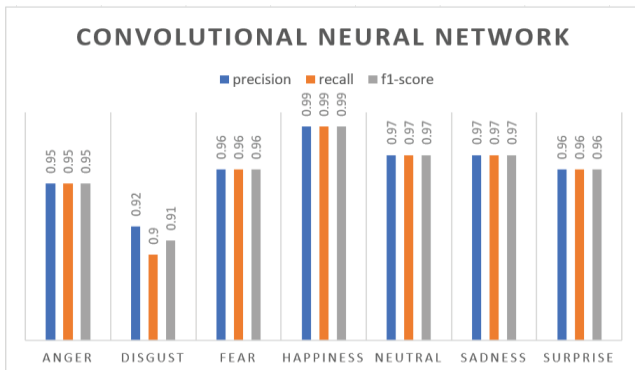


Fig. 2. Performance analysis of CNN model

TABLE IV. PERFORMANCE ANALYSIS OF MOBILENET

	PRECISION	RECALL	F1-SCORE	SUPPORT
Angry	0.85	0.75	0.80	100
Disgusted	0.86	0.79	0.82	100
Fearful	0.84	0.76	0.80	100
Happy	0.95	0.98	0.97	100
Neutral	0.73	0.86	0.79	100
Sad	0.84	0.80	0.82	100
Surprised	0.83	0.92	0.87	100
Accuracy			0.84	700
Macro avg	0.85	0.84	0.84	700
Weighted avg	0.85	0.84	0.84	700

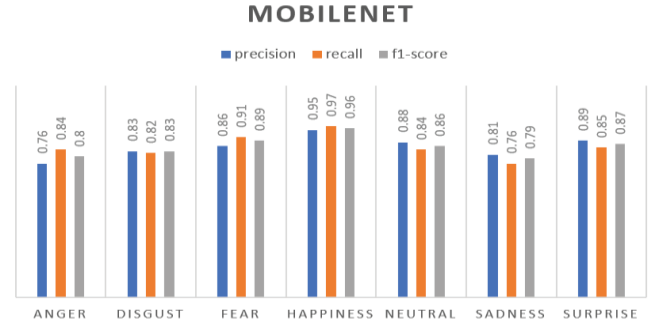


Fig. 3. Performance analysis of Mobilenet model

The performance metrics of various emotions are analyzed using LSTM model. Table 5 shows the performance analysis values and same are depicted in Figure 4.

TABLE V. PERFORMANCE ANALYSIS OF LSTM

	PRECISION	RECALL	F1-SCORE	SUPPORT
Angry	0.85	0.75	0.80	100
Disgusted	0.86	0.79	0.82	100
Fearful	0.84	0.76	0.80	100
Happy	0.95	0.98	0.97	100
Neutral	0.73	0.86	0.79	100
Sad	0.84	0.80	0.82	100
Surprised	0.83	0.92	0.87	100
Accuracy			0.84	700
Macro avg	0.85	0.84	0.84	700
Weighted avg	0.85	0.84	0.84	700

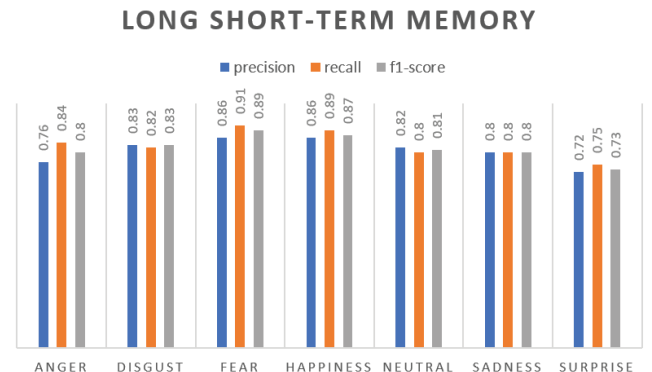


Fig. 4. Performance analysis of LSTM mode

V. CONCLUSION

The field of emotion detection has attracted considerable attention in recent years due to its potential applications in various domains such as healthcare, marketing, and education. Emotion recognition is the process of recognizing and analyzing human emotions using various machine learning techniques. In this context, the comparative analysis of emotion recognition using different methods such as CNN, MobileNet and LSTM has become a key area of research. In a comparative analysis of emotion detection using these methods, CNN was found to have the highest accuracy. CNN was able to detect and recall emotions with a higher level of accuracy compared to MobileNet and LSTM. The results indicate that CNN is a reliable and accurate method for emotion recognition. The accuracy of the proposed models such as CNN, Mobilenet, and LSTM is 97%, 87%, and 82%, respectively. However, it is important to note that the choice of emotion detection method depends on various factors. For example, if real-time detection is required, MobileNet may be a more suitable choice due to its lower computational power requirements. In addition, the type of emotion being detected may also influence the choice of method. Some emotions may be easier to identify using one method than another. A comparative analysis of emotion detection using CNN, MobileNet and LSTM showed that CNN is the most accurate method. However, the choice of method may depend on various factors, and researchers and practitioners should evaluate the pros and cons of each method before making a final decision.

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